Comparing AI in Online Learning: The Transition and Trade-offs Between Intent-Based Learning Assistants and LLM-Chatbots in MOOCs

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Theresa Zobel ; Christoph Meinel All Authors

Abstract:
In the evolving landscape of Massive Open Online Courses (MOOCs), Artificial Intelligence (AI) technologies, notably intent-based learning assistants and Large Language Model (LLM)-Chatbots, present innovative avenues for personalizing and enhancing online education. This paper conducts a succinct comparative analysis of these AI approaches, highlighting their respective strengths and limitations in MOOC environments. Intent-based assistants offer precise, tailored support by responding to specific user intents, facilitating a more structured learning experience. Conversely, LLM-Chatbots, with their vast knowledge bases and conversational capabilities, provide flexible and wide-ranging interactions, encouraging exploratory learning.This analysis further considers the impact of these technologies on fostering social learning communities, addressing ethical and privacy concerns, and their applicability across diverse educational contexts. By comparing the trade-offs between intent-based assistants and LLM-Chatbots, the paper aims to equip educators and technologists with insights into their optimal integration for improving MOOCs. It underscores the importance of aligning AI technologies with educational goals to enhance learning accessibility and quality, especially for underrepresented groups, thereby contributing to the development of more inclusive and effective online learning environments. This research serves as a guide for future AI applications in education, emphasizing the need for a balanced approach in leveraging AI to benefit learners globally.

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I. Introduction

The integration of AI into the educational landscape represents one of the most significant technological advancements of the 21st century, reshaping the way we teach and learn. From personalized learning paths to intelligent tutoring systems, AI has the potential to tailor education to individual needs, fostering deeper understanding and engagement among lea

rticularly impactful in the domain of MOOCs, which have democratized access to education on a global scale. MOOCs, by their very nature, require innovative technological solutions to manage the diverse needs of thousands of learners simultaneously, making